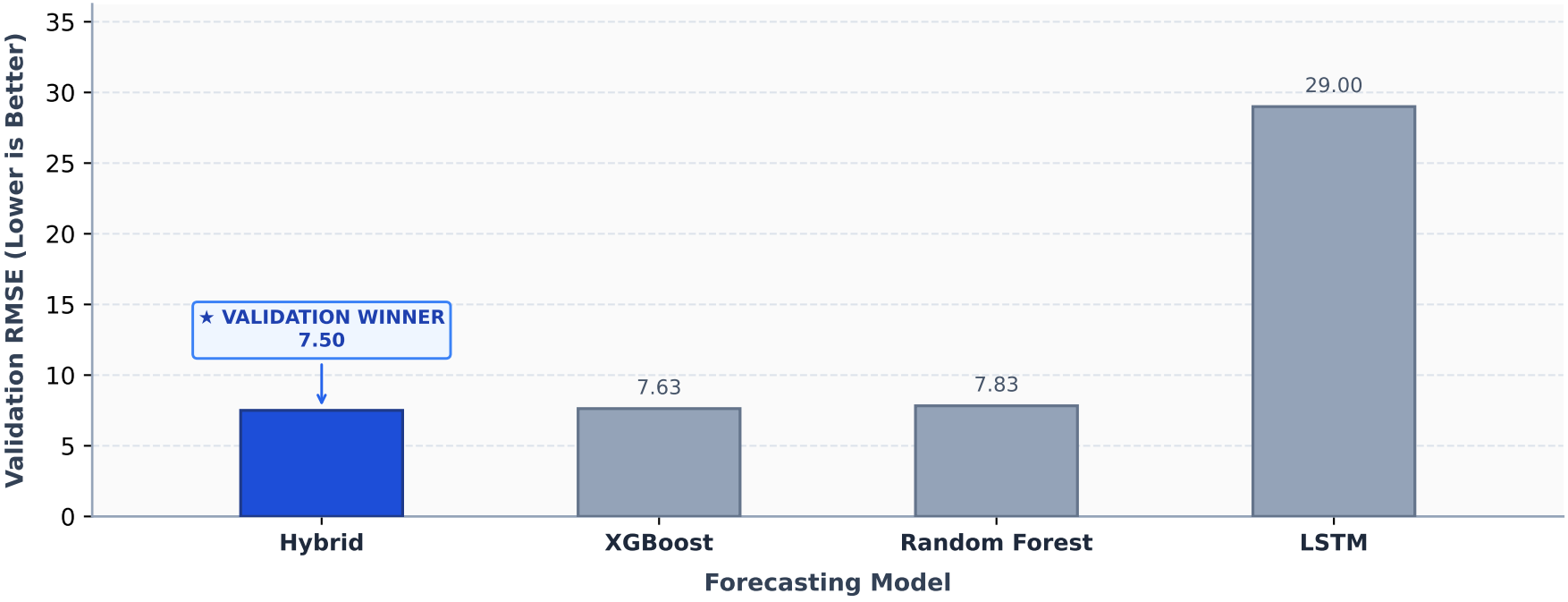
Retail Sales — Validation-Driven Model Benchmark

Model selection based on validation performance

Selected on Validation Set | Criterion: Minimum Validation RMSE

Validation RMSE by Candidate Model



Rank	Model	Val RMSE ↓	Val MAE ↓	Val MAPE ↓	Val R^2 ↑	Test RMSE ↓
#1	**★ Hybrid (Recommended Winner)**	**7.50**	5.55	13.70%	0.9287	23.00
#2	XGBoost	**7.63**	5.71	14.39%	0.9263	22.27
#3	Random Forest	**7.83**	5.71	13.64%	0.9224	23.96
#4	LSTM	**29.00**	25.48	83.92%	-0.0649	45.55

Model selection is performed using validation RMSE. The held-out test set is reserved for final unbiased evaluation.

Selected Model: Hybrid (Validation Winner) | Validation RMSE: 7.50